

# Shuigen Liu

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## Positions

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- 2026 – Now**      **Postdoctoral Fellow**, Pennsylvania State University  
Mentor: Prof. John Harlim
- 2025 – 2026**      **Research Fellow**, National University of Singapore
- 2024 Jul – Dec**   **Visiting student**, Massachusetts Institute of Technology

## Education

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- 2021 – 2025**      **Ph.D. in Mathematics**, National University of Singapore  
Advisors: Prof. Xin T. Tong and Prof. Weizhu Bao
- 2017 – 2021**      **B.S. in Mathematics**, Peking University

## Research Interests

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My research lies at the intersection of computational mathematics, applied probability and statistics. I focus on analysis and development of algorithms for high-dimensional sampling and uncertainty quantification, with applications in Bayesian inference, inverse problems and generative modeling. I am particularly interested in localization methods for sampling. I also work on ensemble Kalman methods for high-dimensional inference and data assimilation.

Keywords: Sampling, uncertainty quantification, localization, dimension reduction, data assimilation.

## Publications and Preprints

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### Preprints

- [6] Georg A. Gottwald, **Shuigen Liu**, Youssef Marzouk, Sebastian Reich, and Xin T. Tong, *Localized diffusion models*, arXiv:2505.04417, submitted to Bernoulli (2026).
- [5] Tiangan Cui, **Shuigen Liu**, and Xin T. Tong, *Stein's method for marginals on large graphical models*, arXiv:2410.11771, submitted to Bernoulli (2026).

### Published/accepted

- [4] **Shuigen Liu**, and Xin T. Tong, *Spectral Gap of Metropolis Algorithms for Non-smooth Distributions under Isoperimetry*, Jpn. J. Stat. Data Sci. (2026).
- [3] Ningxin Liu, **Shuigen Liu**, Xin T. Tong, and Lijian Jiang, *Estimate of Koopman modes and eigenvalues with Kalman filter*, SIAM J. Sci. Comput., 48 (2026), pp. A512–A539.
- [2] Rafael Flock, **Shuigen Liu**, Yiqiu Dong, and Xin T. Tong, *Local MALA-within-Gibbs for Bayesian image deblurring with total variation prior*, SIAM J. Sci. Comput., 47 (2025), pp. A2127–A2153.
- [1] **Shuigen Liu**, Sebastian Reich and Xin T. Tong, *Dropout ensemble Kalman inversion for high dimensional inverse problems*, SIAM J. Numer. Anal., 63 (2025), pp. 685–715.

## Teaching

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### Pennsylvania State University

Lecturer    Calculus With Analytic Geometry I, Fall 2026

### National University of Singapore

TA            Finite Element Method, Spring 2025

TA            Essential Data Analytics Tools: Numerical Computation, Spring 2024

TA            Calculus for Computing, Spring 2023 and Fall 2023

TA            Essential Data Analytics Tools: Convex Optimization, Spring 2023

## Awards

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**2024**    Faculty of Science PhD Conference Award, National University of Singapore

**2024**    Overseas Research Immersion Award, National University of Singapore

**2021**    President's Graduate Fellowship, National University of Singapore

**2021**    Excellent Undergraduate, Peking University

## Service

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Reviewer for: SIAM Journal on Numerical Analysis, SIAM/ASA Journal on Uncertainty Quantification, Journal of Machine Learning Research, Statistics and Computing, Japanese Journal of Statistics and Data Science, Journal of the Royal Statistical Society Series B

## Conferences

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**Jul 2025**    Efficient Sampling Algorithms for Complex Models, IMS NUS, Singapore.

**Jun 2025**    MATRIX Research Program: Bayesian Learning of Very High-Dimensional Physical Process Models, MATRIX, Melbourne, Australia.

**May 2025**    2025 STU-NUS PhD Forum, Shanghai Jiao Tong University, Shanghai, China.

**Oct 2024**    FDS Conference: Recent Advances and Future Directions for Sampling, Yale University, New Haven, U.S.

**Feb 2024**    2024 SIAM Conference on Uncertainty Quantification (UQ24), Trieste, Italy.

**Aug 2023**    2023 International Congress on Industrial and Applied Mathematics (ICIAM 2023), Waseda University, Tokyo, Japan.

**Jun 2023**    Foundations of Computational Mathematics 2023 Conference (FoCM 2023), Sorbonne University, Paris, France.

**Jul 2022**    2022 International Conference on Scientific Computation and Differential Equations (SciCADE 2022), University of Iceland, Reykjavík, Iceland.